

Cloudflare Radar 2024

Year in Review

Internet patterns and trends observed through
Cloudflare's global network during 2024

2024 Annual Report

17% Global Traffic Growth and Starlink's 3.3x Surge

17%

Year-over-year global
Internet traffic growth

3.3x

SpaceX Starlink traffic
surge in 2024

41% of global traffic from mobile devices

28%

IPv6 adoption of dual-stack traffic



Google Leads Top 10 Services as Generative AI Rises

Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

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Generative AI continued its rapid rise throughout 2024, while the Metaverse receded into the background.

Rankings based on Cloudflare 1.1.1.1 DNS resolver data.

Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3



2% →

13%

Post-quantum encrypted TLS 1.3 traffic after Chrome 124 default enablement (Apr 17)

21%

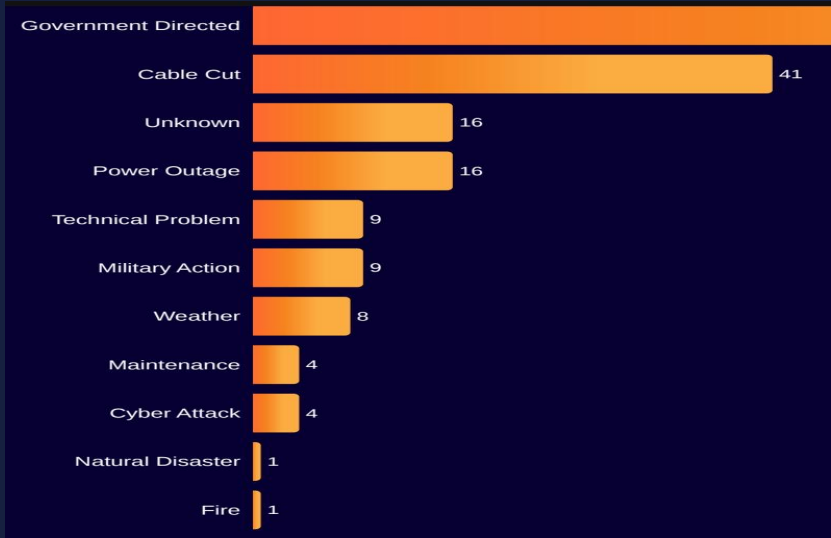
HTTP/3 adoption — faster connections, packet-loss mitigation, and encryption by default via QUIC

Go — Top API client language (~12% of automated API requests) | **Chrome** — Most popular browser globally |
Google Search — 89% of referral traffic

Firefox rolling out post-quantum by default | Safari began initial testing | Node.js, Python (9.6%), Java (7.4%), .NET (3.6%) follow Go in API clients

225 Major Disruptions: Government Shutdowns Lead Outage Causes

Cause	Count
Government Directed	116+
Cable Cut	41
Unknown	16
Power Outage	16
Technical Problem	9
Military Action	9
Weather	8
Maintenance	4
Cyber Attack	4
Natural Disaster	1
Fire	1



Government-directed shutdowns accounted for 116+ of 225 disruptions — over half of all observed outages. Cable cuts (41) and power outages (16) were distant second and third causes.

6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

6.5%

of all traffic mitigated
by Cloudflare

68%

of global bot traffic from
top 10 countries

4.3%

of emails processed
were malicious

Most Attacked Sector

Gambling & Games

Leading industry category targeted by
attacks throughout 2024

Log4j Still Exploited

CVE-2021-44228

Actively targeted 3+ years after
disclosure — underscoring persistent
vulnerability risks

Email Threats

**43% with deceptive
links**

Of malicious emails, 43% contained
deceptive links. .bar TLD originated
the most spam

RPKI valid routes increased by 6.4% — progress toward improved Internet routing security

Five Takeaways Shaping the 2025 Internet Outlook

01

Traffic Growth Trajectory

17% global growth with Starlink's 3.3x surge signals expanding connectivity reach — satellite Internet is closing the access gap.

02

Post-Quantum Momentum

Encryption adoption leapt from 2% to 13% of TLS 1.3 traffic. Browser-driven defaults are accelerating the quantum-ready transition.

03

Connectivity Fragility

Government-directed shutdowns caused 116+ of 225 disruptions. Political actions remain the single largest threat to Internet availability.

04

Persistent Security Threats

6.5% of traffic mitigated, Log4j still exploited after 3 years, and 4.3% of emails were malicious — legacy vulnerabilities endure.

05

Evolving Developer Ecosystem

Go became the top API client language (~12%), HTTP/3 reached 21% adoption — the infrastructure stack is actively modernizing.